Eigenvalues, Eigenvectors, and Positive (Semi)Definite Matrices

Conventions. Vectors are bold (e.g., $\boldsymbol{x}$), matrices use uppercase letters (we use A for matrices), $\boldsymbol{x}^\top$ denotes transpose, and $\|\cdot\|$ is the Euclidean norm. All matrices are real unless stated otherwise.

1 Eigenvalues and Eigenvectors

Definition 1 (Eigenpair)**.** *For a square matrix $A \in \mathbb{R}^{n\times n}$, a scalar $\lambda \in \mathbb{R}$ is an* eigenvalue *and a nonzero vector $\boldsymbol{v} \in \mathbb{R}^n$ is a corresponding* eigenvector *if*

$$A\boldsymbol{v} = \lambda \boldsymbol{v}.$$

Intuitively, A scales $\boldsymbol{v}$ by λ without changing its direction.

How to find them

Rearrange the eigenvalue equation as $(A - \lambda I)\boldsymbol{v} = \mathbf{0}$. Nontrivial solutions exist iff the coefficient matrix is singular:

$$\det(A - \lambda I) = 0.$$

- The roots of $\det(A - \lambda I)$ are the eigenvalues $\{\lambda_i\}_{i=1}^n$.
- For each λ, the eigenvectors form the nullspace $\mathcal{N}(A - \lambda I)$.

Key spectral facts

For any $A \in \mathbb{R}^{n\times n}$:

- $\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$ (sum of eigenvalues).
- $\det(A) = \prod_{i=1}^n \lambda_i$ (product of eigenvalues).
- A is invertible iff 0 is not an eigenvalue.

Theorem 1 (Spectral theorem for real symmetric matrices)**.** *If $A = A^\top$, then A is orthogonally diagonalizable: there exists an orthogonal Q and diagonal $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ with real λ_i such that*

$$A = Q\Lambda Q^\top.$$

Moreover, eigenvectors associated with distinct eigenvalues are orthogonal.

Proposition 1 (Rayleigh quotient)**.** *For symmetric A, the* Rayleigh quotient $R_A(\boldsymbol{x}) = \dfrac{\boldsymbol{x}^\top A\boldsymbol{x}}{\boldsymbol{x}^\top \boldsymbol{x}}$ *satisfies*

$$\lambda_{\min} = \min_{\|\boldsymbol{x}\|=1} \boldsymbol{x}^\top A\boldsymbol{x} \quad \textit{and} \quad \lambda_{\max} = \max_{\|\boldsymbol{x}\|=1} \boldsymbol{x}^\top A\boldsymbol{x}.$$

The optimizers are the eigenvectors for $\lambda_{\min}$ and $\lambda_{\max}$.

2 Positive (Semi)Definite Matrices

Throughout this section we consider *symmetric* matrices $A = A^\top$.

Definition 2 (Quadratic form and definiteness)**.** *For $A = A^\top$ and $\boldsymbol{x} \in \mathbb{R}^n$, the scalar $\boldsymbol{x}^\top A\boldsymbol{x}$ is the* quadratic form*.*

- *A is* positive definite (PD) *if $\boldsymbol{x}^\top A\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \mathbf{0}$.*
- *A is* positive semidefinite (PSD) *if $\boldsymbol{x}^\top A\boldsymbol{x} \geq 0$ for all $\boldsymbol{x}$.*
- *A is* indefinite *if the quadratic form takes both positive and negative values.*

We write $A \succ 0$ for PD and $A \succeq 0$ for PSD.

Eigenvalue characterization

Proposition 2. *Let $A = A^\top$. Then:*

- *$A \succ 0$ iff all eigenvalues are positive: $\lambda_i > 0$.*
- *$A \succeq 0$ iff all eigenvalues are nonnegative: $\lambda_i \geq 0$.*
- *A is indefinite iff it has both positive and negative eigenvalues.*

Equivalent tests and useful properties

- **Sylvester's criterion (PD).** $A \succ 0$ iff all *leading principal minors* are positive:

$$\det A_{1:k,\,1:k} > 0 \quad \text{for } k = 1, 2, \ldots, n.$$

- **Principal minors (PSD).** $A \succeq 0$ iff all principal minors are nonnegative (checking all principal minors is more involved; eigenvalues are often simpler).
- **Factorization test.** $A \succeq 0$ iff there exists B with $A = B^\top B$ (Gram form). If, in addition, B has full column rank, then $A \succ 0$.
- **Cholesky factorization.** $A \succ 0$ iff there is a unique lower-triangular L with positive diagonal such that $A = LL^\top$ (a practical numerical test).
- **Inverse.** If $A \succ 0$, then $A^{-1} \succ 0$.
- **Convex cone.** The set $\mathbb{S}^n_+$ (and $\mathbb{S}^n_{++}$) is a convex cone: if $A, B \succeq 0$ and $\alpha, \beta \geq 0$, then $\alpha A + \beta B \succeq 0$.
- **Optimization link.** If f is twice differentiable and its Hessian $\nabla^2 f(\boldsymbol{x}) \succeq 0$ for all $\boldsymbol{x}$, then f is convex. If $\nabla^2 f(\boldsymbol{x}) \succ 0$, f is strictly convex.

3 Worked Examples (2×2 and 3×3)

We illustrate PSD/PD, quadratic forms, and eigen-analysis on small matrices. Throughout, vectors are bold (e.g., $\boldsymbol{x}$), and matrices are denoted by A_i.

Example 1 (2×2 PD).

$$A_1 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Eigenvalues. For a 2×2 matrix, $\lambda = \dfrac{\operatorname{tr}(A_1) \pm \sqrt{\operatorname{tr}(A_1)^2 - 4\det(A_1)}}{2}$. Here $\operatorname{tr}(A_1) = 4$, $\det(A_1) = 3$, hence

$$\lambda_{1,2} = \frac{4 \pm \sqrt{16-12}}{2} = \frac{4 \pm 2}{2} \in \{3, 1\}.$$

Both are > 0, so A_1 is PD.
Quadratic form. For $\boldsymbol{x} = (x_1, x_2)^\top$,

$$\boldsymbol{x}^\top A_1 \boldsymbol{x} = 2x_1^2 + 2x_2^2 + 2x_1x_2 = (x_1 + x_2)^2 + (x_1^2 + x_2^2) \ > \ 0 \text{ for } \boldsymbol{x} \neq \mathbf{0}.$$

Eigenvectors. For $\lambda = 3$, one eigenvector is $\boldsymbol{v} = [1, 1]^\top$; for $\lambda = 1$, one is $\boldsymbol{v} = [1, -1]^\top$.

Example 2 (2×2 PSD but not PD).

$$A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Eigenvalues. $\operatorname{tr}(A_2) = 2$, $\det(A_2) = 0$. Thus

$$\lambda_{1,2} = \frac{2 \pm \sqrt{4-0}}{2} \in \{2, 0\}.$$

All eigenvalues are ≥ 0, so A_2 is PSD but not PD (zero eigenvalue).
Quadratic form. For $\boldsymbol{x} = (x_1, x_2)^\top$,

$$\boldsymbol{x}^\top A_2 \boldsymbol{x} = x_1^2 + 2x_1x_2 + x_2^2 = (x_1 + x_2)^2 \ \geq \ 0,$$

with equality when $x_1 = -x_2$ (nullspace spanned by $[1, -1]^\top$).
Eigenvectors. For $\lambda = 2$, $\boldsymbol{v} = [1, 1]^\top$; for $\lambda = 0$, $\boldsymbol{v} = [1, -1]^\top$.

Example 3 (Contrast: 2×2 indefinite).

$$A_{\text{indef}} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}.$$

Its eigenvalues are $\pm\sqrt{5}$, one positive and one negative, hence *indefinite*. Indeed, $\boldsymbol{x}^\top A_{\text{indef}} \boldsymbol{x}$ can be negative; for example, with $\boldsymbol{x} = [0, 1]^\top$, $\boldsymbol{x}^\top A_{\text{indef}} \boldsymbol{x} = -1 < 0$.

Example 4 (3×3 PD (tridiagonal)).

$$A_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

This A_3 is symmetric with eigenvalues

$$\lambda_1 \approx 3.4142, \qquad \lambda_2 = 2, \qquad \lambda_3 \approx 0.5858,$$

all strictly positive, so A_3 is PD.
A sample eigenvector. For $\lambda = 2$, one eigenvector is $\boldsymbol{v} = [1,\,0,\,-1]^\top$ since

$$A_3 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

Quadratic form. For any $\boldsymbol{x} = (x_1, x_2, x_3)^\top$,

$$\boldsymbol{x}^\top A_3 \boldsymbol{x} = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 = (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_2^2 \;>\; 0 \text{ for } \boldsymbol{x} \neq \boldsymbol{0}.$$

Example 5 (3×3 PSD via Gram (and a non-PD variant)). Let

$$B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix}, \qquad A_4 = B^\top B = \begin{bmatrix} 10 & 3 & 5 \\ 3 & 1 & 1 \\ 5 & 1 & 5 \end{bmatrix}.$$

By the factorization $A_4 = B^\top B$, the matrix is PSD; its eigenvalues (numerically) are

$$\lambda_1 \approx 13.05, \qquad \lambda_2 \approx 2.36, \qquad \lambda_3 \approx 0.59,$$

so A_4 is in fact PD.

To obtain PSD but not PD, take a rank-deficient factor:

$$C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}, \qquad A_5 = C^\top C = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}.$$

Here $\operatorname{rank}(C) = 1$, hence $\operatorname{rank}(A_5) = 1$ and A_5 is PSD but not PD. The eigenvalues are $\{70,\,0\}$. Moreover,

$$\boldsymbol{x}^\top A_5 \boldsymbol{x} = \|C\boldsymbol{x}\|^2 = (x_1 + 2x_2)^2 + (2x_1 + 4x_2)^2 + (3x_1 + 6x_2)^2 \geq 0,$$

with equality when $x_1 = -2x_2$ (the nullspace).

4 Summary Tests for a Symmetric Matrix A

Test	**Positive Definite (PD)**	**Positive Semidefinite (PSD)**
Eigenvalues	all $\lambda_i > 0$	all $\lambda_i \geq 0$
Quadratic form	$\boldsymbol{x}^\top A\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$	$\boldsymbol{x}^\top A\boldsymbol{x} \geq 0$ for all $\boldsymbol{x}$
Factorization	$A = B^\top B$ with B full column rank	$A = B^\top B$ for some B
Principal minors	all leading principal minors > 0 (Sylvester)	all principal minors ≥ 0

Remark 1. Testing PSD via minors requires checking *all* principal minors and is often impractical; the eigenvalue test or a Gram/Cholesky factorization is typically simpler.

5 Where PD/PSD Matrices Appear (and Why It Matters)

- **Covariance and Gram matrices** ($A = B^\top B$) are PSD by construction; in kernel methods, the kernel matrix must be PSD.
- **Optimization:** A twice-differentiable f is convex iff $\nabla^2 f(\boldsymbol{x}) \succeq 0$ for all $\boldsymbol{x}$; strict convexity corresponds to $\succ 0$.
- **Numerics:** PD matrices admit stable *Cholesky* factorizations, enabling fast linear solves and least-squares.
- **Geometry:** The Rayleigh quotient ties eigenvalues to extremal curvature of the quadratic form on the unit sphere.

Exercises (Short)

1. For $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ (symmetric), show that $A \succ 0$ iff $a > 0$ and $ac - b^2 > 0$ (Sylvester's criterion in 2×2).
2. Let $A = A^\top$ and define $R_A(\boldsymbol{x}) = \dfrac{\boldsymbol{x}^\top A \boldsymbol{x}}{\boldsymbol{x}^\top \boldsymbol{x}}$. Show that the maximum of R_A over $\|\boldsymbol{x}\| = 1$ equals $\lambda_{\max}(A)$, and characterize the maximizers.
3. Prove that if $A = B^\top B$ then $A \succeq 0$. When is $A \succ 0$?
4. For A_3 in the examples, verify directly that all leading principal minors are positive.

Further reading: Any linear algebra text covering the spectral theorem, Sylvester's criterion, and Cholesky factorization (e.g., Horn & Johnson; Trefethen & Bau).